

POST-HARVEST LOSSES IN AGRICULTURAL CROPS

Comprehensive Guide to Causes, Stages, Pre-Harvest & Post-Harvest Management, and Technology

By studying this lesson, you will acquire the competency to:

- Define post-harvest losses in crop cultivation.
- Identify pre-harvest and post-harvest factors affecting losses.
- List the stages at which post-harvest losses occur sequentially.
- Recognize the economic and quality disadvantages caused by post-harvest losses.
- Describe methodologies for managing pre-harvest and post-harvest factors.
- Experiment with appropriate techniques to minimize losses at various stages.
- Explain the benefits derived from preventing post-harvest losses.

It is a common sight in vegetable markets and weekly fairs (pola) to see large quantities of unfit vegetables, fruits, and agricultural produce discarded at the end of daily operations. Similarly, major economic centers such as Dambulla, Meegoda, Veyangoda, Keppetipola, and Thambuttegama, as well as the Manning Market in Pettah, Colombo, witness massive daily disposal of damaged crop harvests.

Post-harvest loss refers to the degradation in quality and quantity of a crop harvest from the moment it is harvested until it reaches the final consumer. Important aspects of these losses include weight reduction, mechanical damage, post-harvest diseases, and senescence (aging).

Global Comparison: While developed countries experience only about **10%** post-harvest loss, developing nations like Sri Lanka suffer vastly higher losses ranging between **20% and 40%**.

Importance of Addressing Post-Harvest Losses

- Post-harvest loss is the primary reason for the wide gap between the farmer's farmgate selling price and the price paid by the consumer to the vendor.
- These losses reduce both the quality and quantity of produce reaching consumers.
- Significant amounts of vital agricultural inputs—including land, labor, and capital—are wasted due to post-harvest losses.

4.1 Factors Affecting Post-Harvest Losses

Ensuring that harvested produce reaches the consumer safely requires attention not only to post-harvest handling activities but also to pre-harvest conditions starting from crop establishment. Factors affecting post-harvest losses are categorized into two main groups:

- **Pre-Harvest Factors**
- **Post-Harvest Factors**

4.1.1 Pre-Harvest Factors

These are factors affecting the crop from establishment until harvest, directly influencing subsequent post-harvest losses:

- **Crop Selection:** Crops and varieties must match regional climates and intended purposes. For example, the *Karuththacolombu* mango thrives and yields high quality in the dry zone, whereas cultivation in the wet zone fails to maintain proper quality standards.
- **Use of Quality Planting Material:** Healthy planting material ensures a healthy crop. Using disease-infected planting material in ginger cultivation, for instance, spreads disease, increases harvest loss, and lowers produce quality.
- **Irrigation:** Water scarcity causes oranges to develop thick rinds and lower juice content, while excessive water or rain causes leafy greens to rot and become fragile. Post-drought over-irrigation in tomatoes and potatoes causes fruit or tuber cracking.
- **Fertilizer Application:** Improper application is detrimental. Applying nitrogen fertilizer close to harvest rots internal potato tuber tissues. Nutrient deficiencies also cause losses; boron deficiency causes lumpy papayas, while calcium imbalances lead to tomato blossom-end rot, reducing shelf life and market value.
- **Pests and Diseases:** Pests active during the growing season damage harvests—such as fungal attacks on chilli and carrots, fruit fly damage in cucurbits (bitter melon), and mite infestations reducing coconut market value.
- **Soil Factors:** Soil texture, structure, and depth must match crop requirements. Tuber crops require light, deep soils; growing carrots in stony, gravelly soils produces irregularly shaped roots.
- **Agricultural Chemicals:** Misuse of growth hormones (e.g., excessive ethylene application in pineapples to stimulate flowering causes fruit drop) or applying pesticides near harvest without observing pre-harvest intervals leaves chemical residues that ruin crop quality.

Climatic Factors

- **Precipitation (Rainfall):** Rain during or near harvest discolors paddy harvests, reducing market value. Excessive moisture makes fruits and vegetables fleshy and prone to mechanical and pest damage, causing leafy greens to bruise and rot.
- **Temperature:** Temperature affects ripening and quality. Low temperatures alter fruit flavor, while excessive heat causes red blotches on tomatoes. Cultivating crops within their optimal temperature ranges enhances quantitative and qualitative value (e.g., papaya thrives in 28°C–35°C).

- **Light:** Cloudy skies reduce the sweetness of oranges. *Willard* mangoes develop their characteristic color only under adequate sunlight; sunlit branches produce sweeter mangoes compared to sour, shaded ones. Potato tubers exposed to sunlight turn green and become inedible.
- **Relative Humidity:** High humidity increases disease susceptibility and reduces flavor in fruits and vegetables, while low humidity causes passion fruit to shrivel.
- **Wind:** Strong winds disrupt pollination, lower yields, cause misshapen fruits, break branches, and drop immature fruits.

Crop Maintenance

Inadequate crop maintenance—such as failing to cover fruits in home gardens or neglecting tea pruning—reduces yields and alters product quality. Pruning and pollination are essential in passion fruit and grape cultivation to ensure high-quality, high-volume yields.

4.1.2 Post-Harvest Factors

The sequence from harvest to consumption involves several steps:

1. Harvesting
2. Handling (Cleaning, Sorting, and Grading)
3. Packaging
4. Transportation
5. Storage
6. Marketing

Factors causing losses during these steps are called post-harvest factors, broadly categorized into:

- **Internal Factors:** Physiological processes such as respiration and transpiration.
- **External Factors:** Adverse weather, pest damage, and improper handling methods.

4.2 Stages Where Post-Harvest Losses Occur

1. During Harvesting

Improper harvesting practices lead to significant losses:

- **Premature Harvesting:** Causes weight loss and reduced color, aroma, and flavor (e.g., mango, banana, tomato).
- **Late Harvesting:** Increases fibrousness in okra and yard-long beans; paddy grains shatter and germinate on the panicle.
- **Harvesting in Direct Sunlight:** Causes wilting in leafy greens, altering their composition.
- **Harvesting in Rain or Dew:** Reduces storage life and increases disease susceptibility (e.g., chilli, grapes, strawberries).
- **Dropping Produce:** Causes bruising, scratching, premature ripening, and disease vulnerability (e.g., papaya, mango).

- **Using Improper Tools:** Damaging citrus branches and fruits by failing to use secateurs.
- **Incorrect Stem Length:** Improper stem cutting in pineapples causes damage.
- **Latex Staining:** Latex spillage causes chemical burns on fruits like mangoes.
- **Field Heat:** Piling harvested produce in the field builds up heat, making fleshy vegetables, fruits, and roots vulnerable to pathogen attack (e.g., cabbage, potatoes, radishes, carrots, grapes).

2. During Cleaning

Inadequate cleaning (failing to remove caustic latex from mangoes and papayas leading to fungal infection) or over-cleaning (abrasively scouring cassava and potatoes, damaging outer surfaces and inviting fungal infection) causes losses. Using dirty water for washing (e.g., carrots, radishes) also introduces diseases.

3. During Sorting and Grading

Failing to sort damaged, pest-infested, or irregularly shaped produce from healthy stock before packing and transport can ruin the entire lot. Export crops must strictly meet grading standards to avoid rejection.

4. During Packaging

Using unsuitable packaging materials, mixing multiple product types in one container, or overfilling packages (such as standard polypropylene sacks used by farmers) causes severe bruising and crushing.

5. During Transportation

Transportation is a major bottleneck in Sri Lanka. Common deficiencies include:

- Using unsuitable open vehicles (tractor trailers, flatbed lorries) exposing produce to sun, rain, and bird droppings.
- Overloading vehicles, causing crushing.
- Rough loading, unloading, and careless driving (sudden braking).
- Passengers sitting or lying on top of produce during transit.
- Long-distance transport over rough roads, and co-transporting produce with building materials or fertilizers.

6. During Storage

Storage facilities in Sri Lanka are inadequate, especially for fruits and vegetables:

- Inappropriate storage temperatures increase respiration, depleting stored food reserves, nutritional value, and weight.
- Poor hygiene leads to infestations by insects (weevils) and rodents (rats, squirrels).
- Poor ventilation and tight stacking cause spoilage.
- Storing mixed produce together (e.g., storing leafy greens with ripe bananas causes greens to yellow prematurely).

7. During Marketing

Displaying fruits and vegetables in open markets exposes them to animals, dust, moisture, wind, and vehicle exhaust, altering their condition. Consumer handling—such as testing firmness with fingernails, squeezing beans, or breaking the tips of okra to check maturity—also causes severe damage.

4.3 Minimizing Post-Harvest Losses

Post-harvest losses can be minimized by properly managing both pre-harvest and post-harvest factors.

4.3.1 Managing Pre-Harvest Factors

- **Crop Selection:** Select region-appropriate crops and varieties (e.g., Kolikuttu, Willard, and Karuththacolombu mangoes in the dry zone) adhering to Department of Agriculture recommendations and matching production goals (e.g., *Isuru* cassava for export, *Kew* pineapple for canning).
- **Site Selection:** Choose land with suitable soil and environmental factors near infrastructure like roads and electricity.
- **Quality Planting Material:** Use pest-free, high-quality planting material to ensure strong crop establishment.
- **Mitigating Climatic Factors:** Utilize greenhouses, net houses, polythene covers, windbreaks, adjusted cropping calendars, and appropriate irrigation or shading to counter adverse rainfall, wind, temperature, humidity, and light conditions.
- **Water Management:** Provide exact water requirements (e.g., drip irrigation for root zones, sprinkler irrigation for frost/dew removal on leafy greens).
- **Fertilizer Management:** Apply recommended fertilizers at correct times. Use sulfate of potash instead of muriate of potash for Solanaceae crops to enhance quality. Integrate organic and chemical fertilizers carefully.
- **Pest Control:** Practice Integrated Pest Management (IPM), maintain cleanliness, use pest-resistant varieties, and employ alternative traps (light traps, pheromone traps, netting, sound) while avoiding chemical misuse.
- **Crop Maintenance:** Practice pruning, thinning of fruits, and fruit bagging to ensure proper structure, light penetration, and protection from pests and weather.

4.3.2 Managing Post-Harvest Factors (Post-Harvest Technology)

Post-harvest technology utilizes scientific concepts and engineering techniques to protect quantity, quality, and value from harvest to consumption.

- **Harvesting Management:** Harvest at optimal maturity indices. Harvesting too early reduces weight, color, aroma, and flavor; harvesting too late increases fiber or shatters grains. Choose appropriate cool times of day, use clean tools, avoid field piling, and apply hot water treatments for mangoes and papayas.
- **Cleaning:** Wash milky fruits with clean water (avoid washing pineapples; use brushes). Clean root crops gently without damaging skins.

- **Sorting and Grading:** Separate damaged or diseased produce and grade by weight, shape, length, and color.
- **Packaging:** Use protective, produce-specific packaging to prevent mechanical damage.
- **Storage:** Maintain precise temperature, relative humidity, and ventilation. Keep storage clean, pest-free, and separate incompatible crops.

Crop	Temperature	Relative Humidity	Storage Life
Mango	12°C	80% – 85%	18 Days
Beans	10°C	90% – 98%	14 Days

- **Transportation:** Use clean, well-ventilated vehicles with low tire pressure and smooth driving. Avoid overloading, mixing produce with other goods, or letting people sit on cargo.
- **Marketing:** Avoid stacking produce deeply to prevent crushing lower layers. Protect from sun, rain, dust, and maintain cool temperatures.

Glossary / පරිභාෂික ශබ්ද මාලාව

English	සිංහල
Pre harvest factors	පෙර අස්වනු සාධක
Post harvest factors	පසු අස්වනු සාධක
Nutrients deficiencies	පෝෂක ඌනතා
Grading	ශ්‍රේණිගත කිරීම
Sorting	තේරීම
Maturity	පරිණත බව
Handling	පරිහරණය
Storing	ගබඩා කිරීම
Maturity index	පරිණත දර්ශක
Post harvest technology	පසු අස්වනු තාක්ෂණය